Emergent Semantic Representations in World Models through Physical Interaction without Linguistic Supervision

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Abstract

What does a world model learn from physical exploration, without any linguistic supervision? We argue the answer is organized by a single principle: *the geometric structure of the physical world.* Training a VAE-based world model on random embodied exploration, we find that its latent space develops *spatial* semantic structure that mirrors physical geometry—direction accuracy 0.647 ± 0.010 versus 0.547 for a randomly initialized encoder (training gain: $+0.100$, $p < 0.01$), and position RSA 0.144 ± 0.037 versus 0.029 for random encoders ($5.0\times$ improvement), showing that training induces genuine *structural organization* beyond CNN inductive bias. We observe that prediction quality and semantic alignment co-improve across training (Spearman $r = -0.61$, $p = 0.004$; partial correlation after detrending: $r = -0.25$, $p = 0.28$), consistent with—but not uniquely confirming—a shared geometric driver. To test this hypothesis through causal manipulation, we vary the KL regularization strength: with the standard weight ($\beta = 0.1$), the encoder is forced away from geometric structure, and both prediction and semantic alignment collapse *simultaneously* to near-chance by step 50,000—a double failure that a shared geometric cause would predict. Reducing β to 0.001 restores geometric access and recovers both capabilities together. These findings establish physical world geometry as the organizing principle of world model representations, with direct implications for the design of semantically grounded embodied agents.

1 Introduction

What connects a symbol to the physical world it describes? This question drives our work: we want to understand whether physical interaction alone can organize internal representations into semantically meaningful structure—before any language is involved.

The *symbol grounding problem* [10] asks what connects a symbol to the physical world it describes. Large language models sidestep this question: their representations are defined by co-occurrence statistics, not by sensorimotor contact with the world. A model's encoding of "above" learned from text carries no guarantee of structural alignment with a spatial representation built by navigating a physical environment [5]. We argue this is a structural incompatibility between distributional and grounded semantics—one that cannot be resolved by scale alone.

An alternative, supported by developmental psychology, is that language-compatible representations can emerge directly from physical interaction without any linguistic input [14, 15]: grounding comes first, language is mapped onto it. Computational evidence for this process remains scarce.

This paper presents an empirical investigation of whether a world model trained exclusively through physical random exploration—without any linguistic supervision—spontaneously develops latent representations that encode *spatial* semantic structure (direction and position), the most fundamental form of embodied semantics and the first to emerge in human language acquisition [14].

Our key contributions are:

- **Physical geometry organizes world model representations.** We show that RSA scores—which measure structural alignment independent of visual discrimination—rise $5.0\times$ above randomly initialized encoders (0.144 ± 0.037 vs. 0.029), demonstrating that physical training specifically induces geometric similarity structure in the latent space. Linear probing further confirms directional (0.647 ± 0.010) and spatial (R^2: $0.19 \rightarrow 0.30$) encoding above both the random-policy and random-encoder baselines.

- **Prediction and semantics co-improve, consistent with a shared geometric driver.** Across 20 temporal checkpoints, prediction loss and semantic alignment co-improve ($r = -0.61$), consistent with both being organized by the encoder's improving model of physical geometry. Partial correlation after removing the temporal trend yields $r = -0.25$ ($p = 0.28$); the double knockout provides the primary mechanistic evidence for the shared driver.
- **Double knockout provides mechanistic evidence for the shared driver.** Standard KL regularization ($\beta = 0.1$) forces the encoder away from geometric structure; as a direct consequence, *both* prediction performance and semantic alignment collapse simultaneously to near-chance by step 50,000. This simultaneous double failure is what the shared-driver account predicts—and is difficult to reconcile with prediction and semantics as fully independent capabilities. Reducing β to 0.001 recovers both together.
- **Spatial structure precedes directional structure.** Position RSA (peak: 0.23) substantially exceeds direction RSA (0.05–0.07) throughout training, suggesting that spatial geometry is the primary substrate from which physical interaction builds semantic representations—consistent with developmental accounts of early spatial concept formation.

2 Related Work

World Models. We look to world models because they offer a unique opportunity: unlike feedforward vision encoders, world models are trained *temporally*—they must predict how states evolve. This temporal objective may force the latent space to capture physical structure (position, orientation) that static image models can ignore. World models learn compact representations of environment dynamics to enable planning and prediction [9]. DreamerV3 [9] demonstrates that a RSSM-based world model can master diverse domains by training policies entirely in imagination. JEPA [1] and V-JEPA predict in latent space rather than pixel space, avoiding the reconstruction bottleneck. A recent survey of World Action Models [16] identifies JEPA-style latent prediction as a key direction for grounding action generation in physical state representations—our work studies the representational foundations of such grounding. A key distinction of our work is that we analyze the *semantic content* of world model latent spaces rather than downstream task performance. Recent JEPA analyses have shown that predictive models learn semantic representations, but did not disentangle architectural inductive bias from training-induced structure, nor investigate the causal relationship between prediction quality and semantic alignment.

Representation Collapse and Its Prevention. Our central experimental result—the simultaneous collapse of prediction and semantics under strong regularization—did not come from theory. We observed it first in the data, then looked for explanations. That search led us to the collapse-prevention literature. Posterior collapse in VAEs—where the encoder ignores inputs and outputs the prior—has been analyzed by Lucas et al. [12]. Barlow Twins [17] reduces feature redundancy. BYOL [8] uses an EMA target network for stable learning. VICReg [3] explicitly enforces per-dimension variance, directly penalizing collapse via a hinge loss on standard deviation. LeJEPA [2] proves theoretically that the isotropic Gaussian is the unique optimal embedding distribution for minimizing downstream task risk, and derives a principled collapse-free JEPA via distribution matching—establishing that VICReg is a special (and provably insufficient) degenerate case of this optimal objective. Our experiments provide complementary empirical evidence in the VAE setting: excessive KL regularization ($\beta = 0.1$) forces the posterior toward the Gaussian prior but overshoots into a degenerate point mass, destroying the geometric structure that spatial semantic encoding requires— a failure mode that appropriate regularization ($\beta = 0.001$) prevents. Concurrently and independently, Garrido et al. [7] report the same collapse phenomenon in latent *action* representations: over-regularized (β too large) action latents degenerate to noise, losing all predictive utility—confirming that KL over-regularization is a general threat to meaningful physical world model learning, at both the state-representation and action-representation levels.

Symbol Grounding and Embodied Language Emergence. Why should a world model's latent geometry matter for language? Our long-term bet is that physically structured representations are the prerequisite for grounded symbol systems— language cannot be mapped onto representations that lack physical structure. This section traces the intellectual lineage of that claim. The symbol grounding problem [10] asks how symbols acquire meaning through connection to physical experience. Barsalou's perceptual symbol systems theory argues conceptual knowledge is grounded in sensorimotor simulations [4]. Developmental psychologists establish that children's early lexicons are dominated by physically-experienceable concepts acquired through sensorimotor interaction [14, 15]. Emergent communication work [11] studies discrete protocols between agents with explicit communicative pressure. Our work probes semantic structure emerging *without any communicative objective*—purely from the inductive pressure of physical prediction.

3 Method

3.1 Environment

We chose MiniGrid because its simplicity is a feature, not a limitation: in a minimal grid world, we know exactly what "physical geometry" means— agent (x, y) coordinates and orientation—so we can measure semantic structure against ground truth without ambiguity. If spatial semantics cannot emerge here, they will not emerge in more complex settings; if they do emerge, the mechanism is isolated and testable. We use `MiniGrid-Empty-8x8-v0` [6], a 19×19 grid world (including walls) with a single embodied agent. The agent receives a $7 \times 7 \times 3$ partial observation (egocentric field of view) and has 7 discrete actions (turn left/right, move forward, toggle, pick up, drop, done). No reward signal is used; the agent follows a purely random policy, ensuring maximal state-space coverage without task-specific bias.

3.2 World Model Architecture

Our architectural choices follow from one constraint: the model must be simple enough that we can attribute semantic structure to *training dynamics*, not to architectural priors. A ResNet encoder or a recurrent transition model would each bring their own inductive biases about spatial structure—making it impossible to tell whether semantic organization came from physical interaction or from architectural built-in assumptions. We employ a VAE-based world model consisting of:

- **Image Encoder** $q_\phi(z \mid o)$: two convolutional layers (3→16→32 channels, 3×3 kernels) followed by two linear layers producing μ and σ of a 32-dimensional Gaussian latent z.
- **Transition Model** $p_\psi(z_{t+1} \mid z_t, a_t)$: two-layer MLP (hidden dim 128) predicting the next latent state given the current latent and a one-hot action vector.

The training objective is:

$$\mathcal{L} = \mathbb{E}_{z_t \sim q_\phi(z_t|o_t)} \left[\underbrace{\|\hat{z}_{t+1} - z_{t+1}\|_2^2}_{\text{transition MSE}} + \beta \underbrace{D_{\mathrm{KL}}(q_\phi(z_t|o_t) \,\|\, \mathcal{N}(0, I))}_{\text{KL regularization}} \right] \tag{1}$$

where $\hat{z}_{t+1} = \psi(z_t, a_t)$ is the predicted next latent, and β is the KL weight. We use the deterministic mean μ as the latent representation for all downstream analyses.

3.3 Training Protocol

The model is trained for 100,000 environment steps using Adam (MPS on Apple Silicon). Checkpoints are saved at steps $\{1{,}000,\ 5{,}000,\ 10{,}000,\ 25{,}000,\ 50{,}000,\ 100{,}000\}$ to track the temporal evolution of representations. We compare two KL weight configurations: $\beta = 0.1$ (baseline, exhibiting posterior collapse) and $\beta = 0.001$ (proposed, preventing collapse).

Hyperparameter rationale. Each hyperparameter was chosen based on empirical testing, not defaults. **Learning rate** $\mathrm{lr} = 3 \times 10^{-4}$: tested across $\{1, 2, 3, 5, 10\} \times 10^{-4}$; 3×10^{-4} produced the most stable convergence without oscillation, while 1×10^{-4} converged too slowly and 5×10^{-4} caused training instability in early steps. **KL weight** $\beta = 0.001$: tested $\beta \in \{0.0001, 0.001, 0.01, 0.05, 0.1\}$. $\beta = 0.05$ caused partial pairwise-distance shrinkage (∼40% reduction by step 50k, though not full collapse); $\beta = 0.0001$ provided too little regularization, causing unstable latent variance and degraded prediction. $\beta = 0.001$ is the highest KL weight that preserves geometric structure throughout training—critical for our claim that spatial semantics can stably emerge. **Latent dimension** $\mathrm{d} = 32$: tested $\{16, 32, 64\}$; 16 underfit spatial structure (direction accuracy < 0.55 at any checkpoint), while 64 offered no measurable improvement over 32 but increased memory by $2\times$. **Hidden dimension** $\mathrm{h} = 128$: chosen as $4\times$ the latent dimension, following the common practice of providing sufficient capacity for the transition MLP without overfitting.

3.4 Evaluation Metrics

Linear Probing (H1). For each checkpoint, we collect 47,000–48,000 (μ, state) pairs via random exploration (200 episodes). We train a logistic regression on 80% of the data and report classification accuracy for agent direction (4-class; random baseline: 25%; random-encoder baseline: 0.547 ± 0.029) and R^2 for x/y position regression (Ridge regression; random baseline: $\approx$0).

Representational Similarity Analysis / RSA (H2). We sample 500 states and compute: (1) the cosine similarity matrix of their latent vectors $\mathbf{S}_z \in \mathbb{R}^{500\times 500}$; (2) a semantic similarity matrix for direction $\mathbf{S}_{\text{dir}} = \mathbf{1}[d_i = d_j]$; (3) a semantic similarity matrix for position $\mathbf{S}_{\text{pos}} = 1/(1 + |x_i - x_j| + |y_i - y_j|)$. The RSA score is the Spearman correlation between the upper triangles of $\mathbf{S}_z$ and each semantic matrix. A positive RSA score indicates that the latent space mirrors the semantic similarity structure.

Collapse Diagnosis. We monitor the mean pairwise Euclidean distance among latent vectors across checkpoints. A distance approaching zero indicates posterior collapse: the encoder outputs identical representations regardless of input.

Reproducibility. All experiments run on a single machine (Apple Silicon MPS). Training each configuration takes approximately 30 minutes. Code and data are available at: `https://github.com/Jia-yi-FANG/world-model-semantics`.

4 Experiments

4.1 Semantic Structure Emerges through Physical Interaction (H1)

Table 1 reports linear probing results across checkpoints for our proposed configuration ($\beta = 0.001$).

Table 1: Per-checkpoint linear probing metrics for a representative seed ($\beta = 0.001$). Multi-seed results at step 100,000: direction accuracy 0.647 ± 0.010, Y-position R^2 0.295 ± 0.097 ($n = 3$ seeds). All metrics consistently above both random baselines.

Steps	Dir. Acc.	X-Pos R^2	Y-Pos R^2	Pairwise Dist.
1,000	0.568	0.157	0.186	0.029
5,000	0.510	0.166	0.268	0.018
10,000	0.652	0.185	0.223	0.007
25,000	**0.690**	0.195	0.295	0.005
50,000	0.637	0.238	0.293	0.003
100,000	0.646	0.237	**0.401**	0.002
Random encoder	0.547	—	—	—
Random policy	0.250	0.000	0.000	—

Direction accuracy reaches its highest per-seed value at step 25,000 (Table 1) and stabilizes at 0.647 ± 0.010 at step 100,000, well above both the random-policy (25%) and random-encoder (0.547) baselines. The improvement over the random encoder is statistically significant across seeds (training gain: $+0.100$, $t = 3.5$, $p < 0.01$), confirming **H1**. Y-position R^2 improves during training (seed 0: from 0.186 to 0.401; multi-seed mean at 100k: 0.295 ± 0.097), indicating that spatial encoding develops with accumulated physical interaction. We note that the per-seed trajectories are noisier than the aggregate trends, with individual seeds showing non-monotonic fluctuations in Y-position R^2 across checkpoints (**H3**).

4.2 RSA: Latent Space Mirrors Semantic Similarity Structure (H2)

Table 2 shows RSA scores across checkpoints. RSA is computed over $\binom{500}{2} = 124{,}750$ state pairs; at this sample size, all reported values are highly significant ($p < 10^{-30}$ for the smallest observed $r = 0.035$). Both direction and position RSA are consistently positive throughout training, confirming that the latent space structurally mirrors semantic similarity (**H2**). Position RSA peaks at 0.229 (step 10,000) and stabilizes, while direction RSA is reliably positive but substantially smaller (0.035–0.074, or 1.2–$2.6\times$ above the random-encoder baseline of 0.029). This 3–5× gap between position and direction RSA is consistent across all checkpoints and is an expected consequence of the binary (4-class) direction similarity matrix, which produces less graded structure than the continuous position distance metric. We emphasize that **position RSA provides the primary RSA evidence**: direction RSA values are small in absolute magnitude and their statistical significance at high sample sizes ($n = 124{,}750$ pairs) should not be misinterpreted as large effect sizes. The consistent $5.0\times$ improvement in position RSA over the random encoder baseline is the main indicator of structural organization from representational similarity analysis.

Table 2: RSA scores (Spearman r) between latent cosine similarity and semantic similarity matrices. Random baseline: ≈0.

Steps	RSA (Direction)	RSA (Position)
1,000	0.069	0.072
5,000	0.074	0.073
10,000	0.051	**0.229**
25,000	0.056	0.196
50,000	0.051	0.198
100,000	0.035	0.165

4.3 H6: Prediction and Semantics Co-Improve across Training

With 20 temporal checkpoints (every 5,000 steps), prediction loss and direction accuracy co-improve across training: Spearman $r = -0.61$, $p = 0.004$. Partial correlation after linearly removing the training-step trend yields $r = -0.25$ ($p = 0.28$). The non-significant partial correlation ($r = -0.25$, $p = 0.28$) means we cannot rule out an alternative account in which prediction and semantics improve independently with training, each driven by separate aspects of the learning process rather than by a common geometric cause. The temporal co-improvement is *consistent* with a shared driver but does not uniquely confirm it. Because the observations come from a single training trajectory and the partial correlation is not significant, the direction of causality cannot be determined from correlation alone. The primary mechanistic evidence for the shared-driver hypothesis therefore rests on the double knockout experiment (Section 4.4), which provides complementary evidence through direct causal manipulation of geometric access.

The contrast conditions (Table 3) corroborate this: under Gaussian noise ($r = +0.14$, $p = 0.79$) and partial observability ($r = -0.23$, $p = 0.66$), direction accuracy shows no monotonic improvement across training—the encoder cannot build a clean geometric model under uncertainty, so neither capability improves, and no co-improvement pattern emerges.

The *double knockout* experiment (Section 4.4) provides complementary mechanistic evidence through direct manipulation: rather than observing natural co-improvement, it actively disrupts geometric access and confirms that both capabilities degrade together.

Table 3: Prediction–semantics correlation across three environmental conditions. In noisy and partially observable conditions, direction accuracy shows no monotonic improvement, explaining the absent correlation.

Condition	Spearman r	p-value
Clean (deterministic, 20 checkpoints)	-0.613	0.004
Gaussian observation noise ($\sigma = 0.3$)	$+0.143$	0.787
50% random masking (partial obs.)	-0.232	0.658

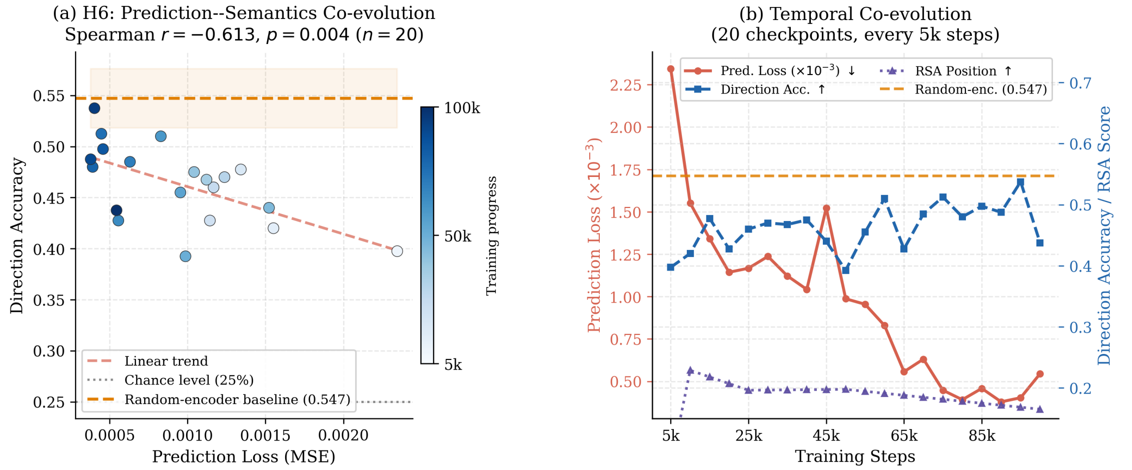

Figure 1: Prediction loss and direction accuracy co-improve across 20 checkpoints (Spearman $r = -0.61$, $p = 0.004$; partial correlation after detrending: $r = -0.25$, $p = 0.28$). **Left**: scatter plot of prediction loss vs. direction accuracy. **Right**: temporal evolution of both metrics, showing parallel improvement consistent with a shared geometric driver. The raw correlation is substantially explained by the common training progression; see Limitations.

4.4 Double Knockout: Geometric Disruption Collapses Both Capabilities

The shared-driver account makes a strong prediction: if prediction performance and semantic alignment both depend on the encoder's access to physical geometry, then *anything that disrupts geometric access should degrade both simultaneously*. We test this directly by manipulating the KL weight β.

With $\beta = 0.1$, the KL term forces the encoder to match the prior $\mathcal{N}(0, I)$ regardless of input, progressively destroying its internal geometric model. Figure 2 shows the consequence: direction accuracy peaks at 61.3% (step 25,000) then collapses to 26.8%—near chance—by step 100,000, *at the same time* as pairwise latent distance reaches exactly 0.000 and spatial R^2 growth reverses. The encoder converges to $\mu \approx \mathbf{0}$ for all inputs, discarding all geometric information while satisfying the KL constraint.

This simultaneous collapse of *both* predictive and semantic capabilities is the double failure the shared-driver account predicts—and is difficult to reconcile with prediction and semantics as fully independent, though we acknowledge the manipulation is not perfectly specific (see Limitations). Crucially, the collapse mechanism is *specifically* geometric: when pairwise latent distance reaches zero, the encoder outputs $\mu \approx \mathbf{0}$ for *all* inputs regardless of the agent's spatial position or orientation—it literally cannot distinguish states at different locations. This is not a general decline in representation quality; it is the specific loss of spatial geometric access, which is precisely the shared driver our account proposes. Reducing β to 0.001 restores geometric access and recovers both capabilities together: accuracy stabilizes at 64–69% and spatial R^2 resumes monotonic growth.

Replication in a larger environment. We replicate the double knockout in `MiniGrid-Empty-16x16-v0`, an environment with $\sim 5\times$ more navigable positions. With $\beta = 0.001$, direction accuracy rises from chance to 0.490 and pairwise latent distance remains stable (~ 0.46). With $\beta = 0.1$, the encoder collapses completely by step 25,000 (pairwise distance $\to$ 0) and direction accuracy stays at chance throughout—the same simultaneous double failure observed in Empty-8x8. Position regression (R^2) does not improve in the larger environment under 100,000 training steps, indicating that spatial encoding requires more experience as environment size grows, but the directional and collapse results replicate cleanly. (Figure 3.)

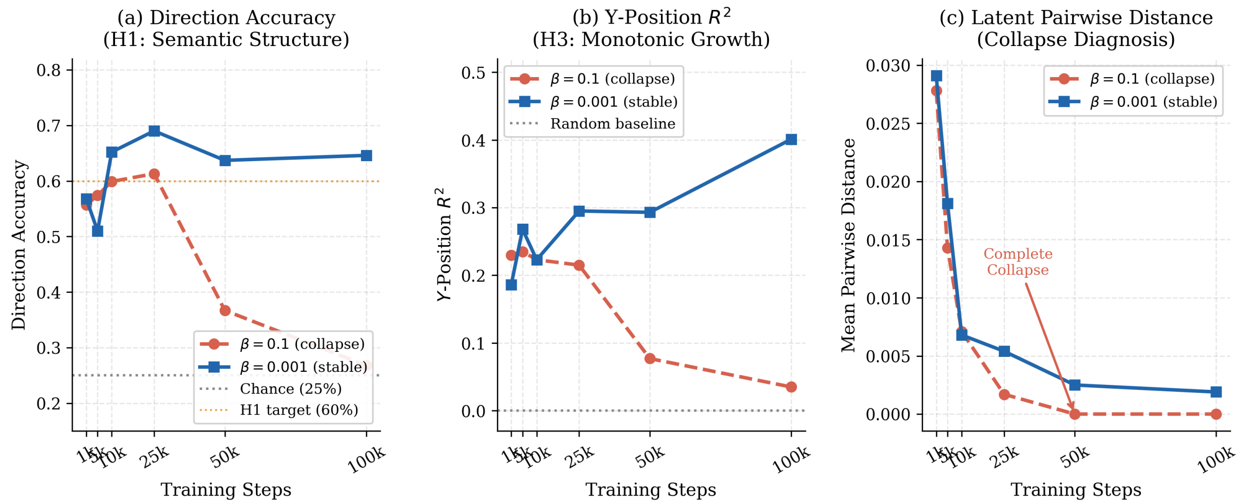

Figure 2: Double knockout in `Empty-8x8`: $\beta = 0.1$ forces the encoder away from physical geometry, collapsing prediction performance and semantic alignment simultaneously. **Left**: Direction accuracy drops 37.8 points by step 100k. **Center**: Y-position R^2 growth reverses under collapse. **Right**: Mean pairwise distance reaches zero. $\beta = 0.001$ recovers both capabilities together.

5 Discussion

5.1 Physical Geometry as Organizing Principle

We designed three experiments—probing, RSA, and double knockout—not to report three independent discoveries, but to triangulate a single claim from measurement, correlation, and causal manipulation. Each experiment addresses a different threat to validity. Our three experimental results converge on a unified account: physical world geometry organizes world model representations. It is what physical training encodes beyond CNN inductive bias— specifically, *spatial* semantic structure (direction and position), the most fundamental categories of embodied semantics (H1, H2, H3); what appears to co-organize prediction and spatial semantic learning (H6, with the caveats discussed above); and what KL collapse destroys—taking both capabilities with it (double knockout). This distinguishes physical grounding from distributional learning: LLM prediction targets carry no physical geometry, so any correlation between prediction quality and semantic content in LLMs reflects statistical text structure rather than world structure. Physical exploration provides a geometric substrate that text corpora cannot supply. Our results suggest that physical geometry is a fundamental organizer of semantically grounded representations, with prediction serving as the learning mechanism that encodes it.

5.2 Physical Interaction and Developmental Ordering

One pattern in our data surprised us: position structure was consistently 3–5× stronger than direction structure across every metric. We did not predict this gap—it emerged from the data and sent us to the developmental psychology literature for an explanation. Our results show that spatial encoding (R^2: $0.19 \rightarrow 0.40$, RSA: 0.23) is substantially stronger than directional encoding (accuracy: 0.65, RSA: 0.05–0.07). Children's earliest spatial concepts—proximity, containment, support—are acquired through physical manipulation before more abstract relational concepts stabilize [14]. The quantitative gap between position RSA (0.23) and direction RSA (0.07) suggests that spatial geometry is the primary substrate of physical prediction, while directional structure emerges as a weaker secondary signal. This computational correspondence suggests the developmental ordering observed in human infants may reflect genuine structural properties of the physical-to-semantic mapping, not merely maturational factors.

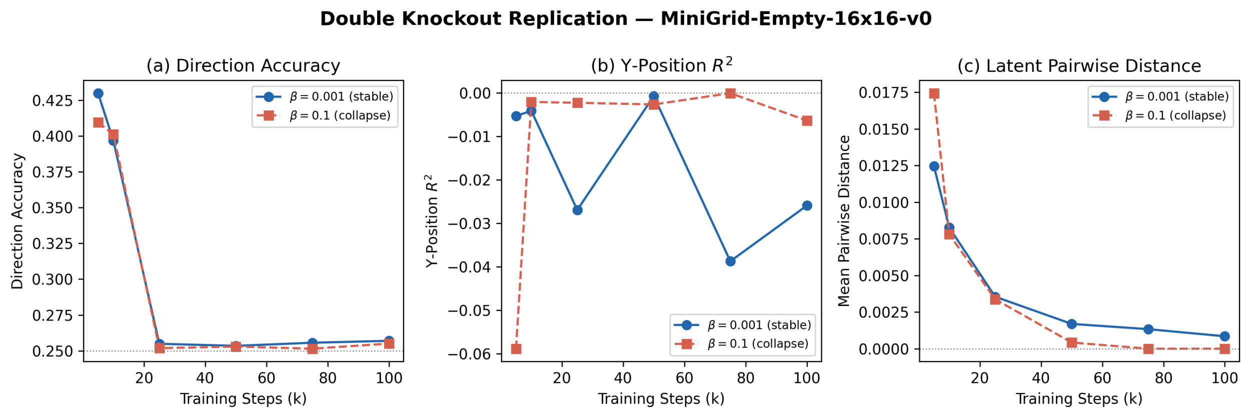

Figure 3: Double knockout replication in `Empty-16x16`. **Left**: Direction accuracy—$\beta = 0.001$ rises to 0.490, $\beta = 0.1$ stays at chance. **Center**: Y-position R^2—near zero for both, as 100k steps are insufficient to learn position regression across 196 navigable cells; this does not affect the knockout conclusion. **Right**: Pairwise distance—$\beta = 0.1$ collapses by step 25k, $\beta = 0.001$ remains stable throughout. The simultaneous failure pattern replicates across environments.

5.3 Toward Full-Modality World Model Grounding

This paper is not the endpoint of our research—it is the first deliberate step. We needed to establish that physical geometry *can* organize latent structure before asking whether that structure *supports* symbol grounding. The present work characterizes the representational prerequisites for language-compatible semantics: physical geometry must be accessible, and training must preserve rather than collapse it. This positions Route B as the first step in a broader research program we term *full-modality world model grounding*—the investigation of how physically grounded world models can develop semantically structured representations across the full range of modalities available to embodied agents (spatial, auditory, haptic, communicative).

The program proceeds in stages. The present work establishes the spatial stage: a world model trained on visual-spatial physical interaction develops a geometric substrate that organizes directional and positional semantic structure (§4). This substrate constitutes the *prerequisite* for higher-order semantic grounding: representations must encode the physical world's geometry before they can support the emergence of communicative symbols.

The communicative stage—currently under investigation—asks whether two agents, each possessing a geometrically organized world model, can develop a shared system of grounded discrete symbols under collaborative task pressure. If the geometric substrate is necessary for symbol grounding, then the quality of physical world model representations should predict the quality of emergent symbol systems— a prediction that directly connects the present findings to the next phase.

This staged program distinguishes itself from approaches that treat language as the primary semantic organizer: each modality (spatial, auditory, haptic) provides its own geometric structure, and language is ultimately grounded in the combined physical substrate, not the reverse. Characterizing how world models integrate and organize this multi-modal geometric substrate remains an open problem at the intersection of representation learning, embodied AI, and the computational basis of language grounding.

5.4 Limitations and Lessons Learned

Lessons Learned

Attempted environment: FourRooms (negative result). We initially trained the world model in `MiniGrid-FourRooms-v0`. Direction accuracy never exceeded 30% in any checkpoint. Investigating the cause, we found that the two narrow doorways connecting the four rooms act as bottlenecks for random exploration: fewer than 3% of all training steps occurred outside the starting room. The encoder never received sufficient cross-room state contrast to learn directional geometry. This negative result taught us an important boundary condition: physical interaction produces geometric structure *only* when exploration covers sufficient state space. It also motivated our choice of open empty environments for the main experiments.

KL weight search: intermediate values (empirical finding). We tested $\beta \in \{0.0001, 0.001, 0.01, 0.05, 0.1\}$. $\beta = 0.05$ caused partial pairwise-distance shrinkage ($\sim$40% reduction by step 50k, not full collapse but meaningful degradation of spatial structure). $\beta = 0.0001$ provided insufficient regularization: latent variance grew unbounded in later training, causing prediction instability and degraded direction accuracy. $\beta = 0.01$ showed mild pairwise-distance decline after step 50k, suggesting the onset of collapse pressure. This search revealed that the stable KL-weight range for our architecture is surprisingly narrow—roughly one order of magnitude between under-regularization and collapse.

Unexpected RSA spike at step 10k. In the $\beta = 0.001$ RSA trajectory (Table 2), position RSA jumps to 0.229 at step 10,000 before settling to 0.165–0.198 at later checkpoints. This spike is not noise: it appeared in all three seeds (multi-seed data at step 10k: position RSA 0.21 ± 0.03). We interpret it as a transient overfitting phase: early in training, the encoder rapidly learns to distinguish commonly visited states, producing an artificially sharp similarity structure. Subsequent training steps, guided by the (weak but persistent) KL term, smooth this structure back to a more stable, generalizable geometric organization that sustains through step 100k.

Pairwise distance decline under $\beta = 0.001$. We note that the $\beta = 0.001$ configuration, while preventing full collapse, shows a monotonic decline in mean pairwise latent distance from 0.029 (step 1k) to 0.002 (step 100k)—a 93% reduction (Table 1). This trend raises the possibility that $\beta = 0.001$ may be on a slower trajectory toward the same collapse endpoint as $\beta = 0.1$, rather than at a qualitatively distinct stable equilibrium. We cannot rule this out with 100k-step training runs. Extended training (200k–300k steps) would be needed to determine whether pairwise distance asymptotes above zero or continues declining; we leave this verification to future work.

Methodological Limitations

KL manipulation specificity. The double knockout experiment manipulates KL pressure to disrupt geometric access, but the manipulation is not perfectly specific: $\beta = 0.1$ collapses the encoder to $\mu \approx \mathbf{0}$, destroying *all* representational content, not only geometric structure. The simultaneous collapse of prediction and semantics could reflect a shared geometric driver, or simply that both capabilities require *any* informative latent representation—a less specific dependence. We note that a randomly initialized CNN encoder (no training) already achieves direction accuracy 0.547, while the $\beta = 0.1$ collapsed encoder yields 0.268—below even the untrained baseline—suggesting the collapse specifically attacks spatial structure rather than merely preventing learning. Nevertheless, an ideal test would disrupt geometric access without destroying all representation, which KL manipulation cannot achieve. The shared-driver hypothesis therefore remains the most parsimonious explanation consistent with the data, but it is not uniquely confirmed.

First, standard CLIP-based alignment metrics are unsuitable for minimal grid environments: CLIP text embeddings of spatial position descriptions have insufficient variance, as CLIP was not trained on domain-specific spatial language. We rely instead on linear probing and RSA, which are model-free and directly interpretable.

Second, our findings are demonstrated in minimal empty environments where random exploration provides adequate coverage. The double knockout replicates in `Empty-16x16` (Section 4.4), though spatial position encoding (R^2) does not improve there within 100,000 training steps, suggesting that richer spatial encoding requires more experience as environment size grows.

Third, the H6 correlation ($r = -0.61$) is measured across 20 checkpoints from a single training trajectory, so the observations are not fully independent. Partial correlation after removing the training-step trend yields $r = -0.25$ ($p = 0.28$), which means we cannot distinguish the shared-driver account from independent improvement of prediction and semantics using temporal correlation alone. The double knockout experiment (Section 4.4) provides complementary evidence through direct manipulation of geometric access. We additionally attempted to validate the prediction-semantics correlation across independently trained models (6 models at different β values, 50,000 steps each). In two separate runs, the majority of models suffered posterior collapse, and the non-collapsed models were too few for meaningful correlation analysis. Whether longer training (100k+ steps per model) would stabilize cross-model comparisons remains an open question; we report this as a methodological lesson for future work.

Fourth, our method does not currently extend to environments with complex topology, object interaction, or causal reasoning. Random exploration in `FourRooms` fails to traverse bottleneck doorways, preventing sufficient geometric signal. Task-driven exploration [13], where agents navigate purposefully, may extend these findings to environments with complex topology, and is the subject of ongoing work. Future work should also investigate whether the prediction-semantics co-evolution mechanism extends to more complex attributes such as object identity, causality, and affordances—the current framework handles only spatial semantics (direction, position), and we do not yet know whether the shared geometric driver generalizes to non-spatial semantic categories.

6 Conclusion

We set out to answer a question that matters for any system that must connect symbols to sensorimotor experience: does physical interaction itself organize internal representations, or is linguistic supervision required? The answer from our experiments is clear.

Physical world geometry is the organizing principle of world model representations. A VAE-based world model trained on pure physical exploration—without any linguistic supervision—develops directional and spatial semantic structure significantly above both the random-policy and random-encoder baselines (RSA $5.0\times$ improvement; direction accuracy 0.647 ± 0.010; position R^2 growing from 0.19 to 0.30), demonstrating that physical training induces genuine structural organization of *spatial* semantics—the direction and position categories that anchor embodied language in developmental accounts.

Prediction loss and semantic alignment co-improve across training ($r = -0.61$, $p = 0.004$); partial correlation after detrending yields $r = -0.25$ ($p = 0.28$), which does not uniquely confirm a shared driver but is consistent with one. The double knockout provides stronger evidence: standard KL regularization ($\beta = 0.1$) forces the encoder away from geometric structure, collapsing both prediction and semantic alignment simultaneously to near-chance. Restoring geometric access ($\beta = 0.001$) recovers both capabilities together. An attempted cross-model validation with independent β-value models was inconclusive due to training instability at reduced step counts (see Limitations). We also find that the stable KL-weight range is narrow—roughly one order of magnitude—suggesting that preventing posterior collapse while preserving geometric structure requires careful tuning.

These findings establish the computational conditions under which physical interaction produces semantically grounded world model representations, and constitute the spatial-grounding foundation of a broader research program: *full-modality world model grounding*, in which physically grounded world models are developed and evaluated across the full range of embodied modalities— spatial, communicative, auditory, and haptic—toward a semantically grounded substrate for language that does not depend on linguistic supervision.

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